

YUAN GAO

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Long-term Goal: Robotist, Professor in Robotics
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RESEARCH

As an assistant professor and roboticist, I have published more than 50 papers in top journals and mainstream field conferences, such as Science Robotics, IEEE T-RO, IEEE IoTJ, IEEE T-MECH, IEEE IROS/ICRA, and IEEE RO-MAN (Total citations: 1335, h-index: 17). I am actively involved in community work, serving as a reviewer for IEEE T-RO, IEEE IoTJ, IEEE T-MECH, IEEE RAL, ICRA, and IROS. In addition to my research, I am dedicated to pedagogical activities, having delivered multiple lectures. I have also been invited to give talks at prestigious institutions such as the University of Cambridge, the Royal Institute of Technology, Sun Yat-sen University, and Shenzhen University. Currently I am leading the (Heterogeneous Multi-)Robot Societalization Lab with 30 students.

"How do we understand human society—a scientific method, a professionally designed system, or our innate curiosity?" In the journey of self-growth, we often pause to ponder life's peculiar questions, and like many, I have become a researcher exploring the unknown. Seeking answers is a lengthy process that requires detours, appreciation of the scenery, existential reflection, and hard work. I am particularly intrigued by heterogeneous multi-robot systems for data interaction and human social dynamics, much like an astronomer's fascination with telescopes and distant stars. This passion has driven me to pursue the creation of an extensive, complex, intelligent, and interactive robotic societal system that mirrors human society, thereby enhancing our self-understanding. This goal encompasses robotics learning and control, natural language processing, neuroscience, and computational psychology, with a keen interest in large language models and neural methods for machine and human behavior modeling. These approaches not only aid in understanding ourselves but also lay the foundation for an adaptive, efficient, and robust complex heterogeneous robot system.

GRANTS

National Key R&D Program of China <i>Human-machine Interaction Software and Hardware System for Hybrid Intelligence</i> Sub-topic Leader / Organization Leader	1,000,000 RMB 2025–2028
Shenzhen Peacock Team Project <i>Mobile Collaborative Robot and Multi-robot Collaboration General System Key Technologies</i> Co-Principal Investigator	1,040,000 RMB 2023–2028
Shenzhen Key R&D Program <i>City-scale Embodied Large Model-driven Swarm Intelligence Collaboration Key Technology R&D</i> Principal Investigator	3,000,000 RMB 2025–2028
Guangdong Province Project <i>Large-scale Intelligent Heterogeneous Multi-robot System based on Graph Representation Multi-agent Reinforcement Learning</i> Principal Investigator	100,000 RMB 2023–2025
Cross-disciplinary Research Fund <i>Ubiquitous Multimodal Data Collection and Analysis for Human Development Study with AI-Driven Heterogeneous Multi-Agent System</i> Co-Principal Investigator	1,500,000 RMB 2025–2028
Cross-disciplinary Research Fund <i>Adaptive Interaction Intervention and Human Development Skill Assessment based on Embodied Multi-Agent System</i> Co-Principal Investigator	1,500,000 RMB 2025–2028
Guangdong Province Project <i>Cloud-Edge-End Asynchronous Cognition and Collaborative Decision-making for Heterogeneous Multi-Agent Systems based on Large Language Models</i>	500,000 RMB

Principal Investigator 2025–2028

Qualcomm – Industry Collaboration Project 1,200,000 RMB

Edge Intelligence and Distributed Cognition in Embodied Multi-Agent Systems

Principal Investigator 2025–2026

Other Projects 4,000,000 RMB

Multi-Scenario Heterogeneous Intelligent Inspection and Three-Dimensional Modeling System for Longwanggou Coal Mine. Industry collaboration with Ordos Guoyuan Mining Development Co., Ltd. (China Datang Group), **Principal Investigator**, (2026–Present).

Autonomous Decision-Making Robot Systems under Communication-Constrained Environments. Collaborative research project with China Electronics Technology Group Corporation (CETC), **Technical Lead**, (2026–Present).

5G-based Heterogeneous Multi-robot Systems Under Uncertain Environments. National Key R&D Program of China on Intelligent Robots, **Main Contributor**, Key special projects of the Ministry of Science and Technology of China, (2020-2023, 2020YFB1313300).

[ANIMATAS](#) Europe Horizon 2020 project, **Main Contributor**, (2019-2022)

Co-adaptive Human-robot Interactive Systems ([COIN](#)) (2016 -), **Main Contributor**, (2018-2021)

AWARDS

Shenzhen Robot Association's Typical Case Award for Robot Applications (2024), Gaogong Golden Ball Award for Innovation Design (2024)

INTERESTS

Embodied-AI based multi-robot systems, Machine behaviour analysis, Robot learning, Deep learning models, Recurrent neural network, Topological interpretation of neural network.

EDUCATION

Visiting Scholar in Prof. Hatice Gunes's team from May to October, 2019 at University of Cambridge

PhD degree in Computer Science at Uppsala University under supervision of Prof. Ginevra Castellano and Prof. Danica Kragic, IEEE Fellow, 09/2016–01/2021

Master's degree at University of Helsinki, **MSc degree in Algorithm and Machine Learning (Sub-majored in Mathematics)** 09/2013–09/2015

Received Bachelor's degree at Vaasa University of Applied Sciences, **BE degree in Embedded Systems (Sub-majored in Telecommunication)**, 09/2009–05/2013

High School Diploma, Huanggang Senior High School, Huanggang, Hubei, China.

EXPERIENCE

Adjunct Assistant Professor & Independent PI

Chinese University of Hong Kong, School of Science and Engineering, Shenzhen 06/2023–Present

Shenzhen Institute of Artificial Intelligence and Robotics for Society, Shenzhen, China 01/2021–Present

- Teach embodied-AI related master-level courses.
- Lead a student research team capable of publishing top-level papers.
- Organized and drove the establishment of the Guangdong Provincial Embodied-AI Innovation Center and the Guangdong Provincial Embodied-AI Engineering Center.

PI of Heterogeneous Multi-Robot Systems Division

Shenzhen Institute of Artificial Intelligence and Robotics for Society, Shenzhen, China

03/2024–Present

- Architect heterogeneous multi-robot systems using multi-modal LLM-based multi-agent framework.

Co-founder & Chief Scientist

Symbiotic-X Co., Ltd., Shenzhen, Qianhai, China

09/2024–Present

- Started an embodied-AI based heterogeneous multi-robot system company as the chief scientist. ([Material of the Company](#))

Deputy Secretary-General

Guangdong Artificial Intelligence and Robotics Association

2026–Present

Past Roles:

Robot Project Researcher

Nokia Research Lab, Espoo, Finland

09/2014–07/2015

Research Assistant

Helsinki Institute of Information Technology,

Helsinki, Finland

05/2014–12/2014

PUBLICATIONS

* First / Co-first Author † Corresponding Author

[1] Xi Chen*(co-first), **Yuan Gao***(co-first), et al., “Beyond Form and Function: Cross-Robot Imitation Learning through Intention Alignment”, Science Robotics, 2026. (Science Robotics 2026, CAS Zone 1 Top)

[2] Tianqiang Yan, **Yuan Gao**†, et al., “ReBeCA: Unveiling Interpretable Behavior Hierarchy behind the Iterative Self-Reflection of Language Models with Causal Analysis”, Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, 2026. (EMNLP 2026)

[3] Yapeng Liu*(co-first), Haonan Zhang*(co-first), Yuming Liu, Lin Zhang, Chen Liang, **Yuan Gao**†, et al., “TeamWeaver: Hybrid LLM and Optimization-based Planning with Transparent Constraints in Heterogeneous Multi-Robot Teams”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026, 1-8. (IROS 2026, Top Robotics Conference)

[4] Haocheng Wang, Lin Wang, Tin Lun Lam, Jianwang Zhai, Dahai Lin, Hexiang Zheng, Xvchun He, **Yuan Gao**†, et al., “Emergent Co-Adaptive Strategies in Heterogeneous Multi-Robot Systems via Meta-Learning”, IEEE International Conference on Robotics and Automation, 2026. (ICRA 2026, Top Robotics Conference)

[5] Jianlin Wang*(co-first), **Yuan Gao***(co-first), et al., “Shape-Adaptive Inspection Robot for Misaligned Interstices in Stacked Logs: Geometry-Guided Design and Experimental Validation”, Advanced Robotics Research, 2026. (ARR 2026)

[6] Ping Feng, **Yuan Gao**†, et al., “OC-HMAS: Dynamic Self-Organization and Self-Correction in Heterogeneous Multi-Agent Systems Using Multi-Modal Large Models”, IEEE Internet of Things Journal, 2025, 1-8. (IoTJ 2025, CAS Zone 1 Top)

[7] Jinlong Shi, **Yuan Gao**, et al., “VLONS: A Vision-and-Language On-device Navigation System with Multimodal Fusion and Modular Framework”, IEEE Transactions on Consumer Electronics, 2025. (TCE 2025)

- [8] Jiaqi Hu, **Yuan Gao**, et al., “Unlocking Drone Perception in Low AGL Heights: Progressive Semi-Supervised Learning for Ground-to-Aerial Perception Knowledge Transfer”, IEEE Transactions on Intelligent Transportation Systems, 2025. (T-ITS 2025, CAS Zone 1 Top)
- [9] Tianqiang Yan, **Yuan Gao**[†], et al., “Entrospect: Information-Theoretic Self-Reflection Elicits Better Response Refinement of Small Language Models”, 2025, 1-8. (ACL 2025, CCF-A)
- [10] Changheng Cai, **Yuan Gao**[†], et al., “Multimodal Deformation Estimation of Soft Pneumatic Gripper During Operation”, 2025, 1-8. (IROS 2025, Top Robotics Conference)
- [11] Chongyang Wang, **Yuan Gao**[†], et al., “Understanding Users’ Perceptions and Expectations toward a Social Balloon Robot via an Exploratory Study”, Proceedings of the ACM Symposium on User Interface Software and Technology, 2025. (UIST 2025, CCF-A)
- [12] Yanjie Liu, **Yuan Gao**, et al., “Configuration-Adaptive Visual Relative Localization for Spherical Modular Self-Reconfigurable Robots”, IEEE International Conference on Robotics and Automation, 2025. (ICRA 2025, Top Robotics Conference)
- [13] Jiaqi Wang, **Yuan Gao**, et al., “Transformable Inspection Robot Design and Implementation for Complex Pipeline Environment”, IEEE Robotics and Automation Letters, 2024, 9 (6), 5815-5822. (RAL 2024)
- [14] Chongyang Wang, **Yuan Gao**[†], et al., “PepperPose: Full-Body Pose Estimation with a Companion Robot”, Proceedings of the CHI Conference on Human Factors in Computing Systems, 2024, 1-8. (CHI 2024, CCF-A)
- [15] Wenqiang Lai, **Yuan Gao**[†], et al., “Vision-Language Model-based Physical Reasoning for Robot Liquid Perception”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024. (IROS 2024, Top Robotics Conference)
- [16] Jian Chen, **Yuan Gao**, et al., “Meta-Reinforcement Learning Based Cooperative Surface Inspection of 3D Uncertain Structures Using Multi-Robot Systems”, IEEE International Conference on Robotics and Automation, 2024, 7201-7207. (ICRA 2024, Top Robotics Conference)
- [17] Hao Zhang, **Yuan Gao**, et al., “An Intention Inference Method for the Space Non-Cooperative Target Based on BiGRU-Self Attention”, Advances in Space Research, 2023, 72 (5), 1815-1828. (Adv. Space Res. 2023)
- [18] Yanlin Wang, **Yuan Gao**, et al., “Asymptotically Efficient Estimator for Range-Based Robot Relative Localization”, IEEE/ASME Transactions on Mechatronics, 2023, 28 (6), 3525-3536. (T-MECH 2023, CAS Zone 1 Top)
- [19] **Yuan Gao**^{*}(first), et al., “Asymmetric Self-Play Enabled Intelligent Heterogeneous Multi-Robot Catching System Using Deep Multi-Agent Reinforcement Learning”, IEEE Transactions on Robotics, 2023, 39 (4), 2603-2622. (T-RO 2023, CAS Zone 1 Top)
- [20] **Yuan Gao**^{*}(first), et al., “Development of an Automated System for the Soldering of USB Cables”, Robotics and Computer-Integrated Manufacturing, 2023, 79, 102440. (RCIM 2023, CAS Zone 1 Top)
- [21] Jian Chen, **Yuan Gao**, et al., “MultiRoboLearn: An Open-Source Framework for Multi-Robot Deep Reinforcement Learning”, IEEE International Conference on Robotics and Biomimetics, 2023, 1-6. (ROBIO 2023)
- [22] Jiaqi Hu, **Yuan Gao**, et al., “Boosting Lightweight Depth Estimation via Knowledge Distillation”, International Conference on Knowledge Science, Engineering and Management, 2023, 27-39. (KSEM 2023)
- [23] Chongyang Wang, **Yuan Gao**, et al., “Learn2Agree: Fitting with Multiple Annotators Without Objective Ground Truth”, International Workshop on Trustworthy Machine Learning for Healthcare, 2023, 147-162. (TML4H Workshop 2023)
- [24] Mian Irtiza Ahmad, **Yuan Gao**, et al., “Emotion and Memory Model for Social Robots: A Reinforcement Learning Based Behaviour Selection”, Behaviour & Information Technology, 2022, 41 (15), 3210-3236. (Behav. Inf. Technol. 2022)
- [25] Jiacheng Tang, **Yuan Gao**, et al., “Learning to Coordinate for a Worker-Station Multi-Robot System in Planar Coverage Tasks”, IEEE Robotics and Automation Letters, 2022, 7 (4), 12315-12322. (RAL 2022)
- [26] Sofia Ahlberg, **Yuan Gao**, et al., “Co-Adaptive Human–Robot Cooperation: Summary and Challenges”, Unmanned Systems, 2022, 10 (02), 187-203. (Unmanned Syst. 2022)

- [27] Xinyue Chen, **Yuan Gao**, et al., “Lapo: Latent-Variable Advantage-Weighted Policy Optimization for Offline Reinforcement Learning”, *Advances in Neural Information Processing Systems*, 2022, 35, 36902-36913. (NeurIPS 2022, CCF-A)
- [28] Hao Guan, **Yuan Gao**, et al., “AB-Mapper: Attention and BiCNet Based Multi-Agent Path Planning for Dynamic Environment”, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022. (IROS 2022, Top Robotics Conference)
- [29] Fuqin Deng, **Yuan Gao**, et al., “Abnormal Occupancy Grid Map Recognition Using Attention Network”, *IEEE International Conference on Robotics and Automation*, 2022, 8666-8672. (ICRA 2022, Top Robotics Conference)
- [30] Jiaqi Hu, **Yuan Gao**, et al., “Progressive Self-Distillation for Ground-to-Aerial Perception Knowledge Transfer”, *arXiv preprint*, 2022, arXiv:2208.13404. (arXiv 2022)
- [31] Mengxi Peng, **Yuan Gao**, et al., “Multilevel Hierarchical Network with Multiscale Sampling for Video Question Answering”, *arXiv preprint*, 2022, arXiv:2205.04061. (arXiv 2022)
- [32] Fangkai Yang, **Yuan Gao**, et al., “A Dataset of Human and Robot Approach Behaviors into Small Free-Standing Conversational Groups”, *PloS One*, 2021, 16 (2), e0247364. (PLoS ONE 2021)
- [33] Fuqin Deng, **Yuan Gao**, et al., “FEANet: Feature-Enhanced Attention Network for RGB-Thermal Real-Time Semantic Segmentation”, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2021. (IROS 2021, Top Robotics Conference)
- [34] Chongyang Wang, **Yuan Gao**, et al., “Leveraging Activity Recognition to Enable Protective Behavior Detection in Continuous Data”, *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2021. (IMWUT 2021, CCF-A)
- [35] Hao Guan, **Yuan Gao**, et al., “AB-Mapper: Attention and BiCNet Based Multi-Agent Path Finding for Dynamic Crowded Environment”, *arXiv preprint*, 2021, arXiv:2110.00760. (arXiv 2021)
- [36] Jian Chen, **Yuan Gao**, et al., “Meta Reinforcement Learning Based Sensor Scanning in 3D Uncertain Environments for Heterogeneous Multi-Robot Systems”, *arXiv preprint*, 2021, arXiv:2109.13617. (arXiv 2021)
- [37] Mengxi Peng, **Yuan Gao**, et al., “Temporal Pyramid Transformer with Multimodal Interaction for Video Question Answering”, *arXiv preprint*, 2021, arXiv:2109.04735. (arXiv 2021)
- [38] Chongyang Wang, **Yuan Gao**, et al., “AgreementLearning: An End-to-End Framework for Learning with Multiple Annotators Without Groundtruth”, *arXiv preprint*, 2021, arXiv:2109.03596. (arXiv 2021)
- [39] **Yuan Gao***(first), et al., “Development of an Autonomous Soldering Robot for USB Wires”, *IEEE/ASME International Conference on Advanced Intelligent Mechatronics*, 2020. (AIM 2020)
- [40] Congcong Li, **Yuan Gao**, et al., “Efficient Learning of Socially Aware Robot Approaching Behavior Toward Groups via Meta-Reinforcement Learning”, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2020, 12156-12159. (IROS 2020, Top Robotics Conference)
- [41] Mengxi Peng, **Yuan Gao**, et al., “Recognizing Micro-Expression in Video Clip with Adaptive Key-Frame Mining”, *arXiv preprint*, 2020, arXiv:2009.09179. (arXiv 2020)
- [42] Xinyue Chen, **Yuan Gao**, et al., “Skew-Explore: Learn Faster in Continuous Spaces with Sparse Rewards”, 2020. (OpenReview 2020)
- [43] **Yuan Gao***(first), “Machine Behavior Development and Analysis using Reinforcement Learning”, *Acta Universitatis Upsaliensis*, 2020. (PhD Thesis)
- [44] **Yuan Gao***(first), et al., “Vision-Based Grasping and Manipulation of Flexible USB Wires”, *IEEE International Conference on Robotics and Biomimetics*, 2019, 1083-1088. (ROBIO 2019)
- [45] **Yuan Gao***(first), et al., “Fast Adaptation with Meta-Reinforcement Learning for Trust Modelling in Human-Robot Interaction”, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2019. (IROS 2019, Top Robotics Conference)
- [46] **Yuan Gao***(first), et al., “Learning Socially Appropriate Robot Approaching Behavior Toward Groups Using Deep Reinforcement Learning”, *IEEE International Conference on Robot and Human Interactive Communication*, 2019. (RO-MAN 2019)

- [47] Daniel Hernandez, **Yuan Gao**, et al., “A Generalized Framework for Self-Play Training”, IEEE Conference on Games, 2019, 1-8. (CoG 2019)
- [48] Peng Zhang, **Yuan Gao**, et al., “Bandit Learning with Concurrent Transmissions for Energy-Efficient Flooding in Sensor Networks”, EAI Endorsed Transactions on Industrial Networks and Intelligent Systems, 2018, 4 (13). (EAI INIS 2018)
- [49] Congcong Li, **Yuan Gao**, et al., “Effects of Posture and Embodiment on Social Distance in Human-Agent Interaction in Mixed Reality”, Proceedings of the International Conference on Intelligent Virtual Agents, 2018. (IVA 2018)
- [50] **Yuan Gao**^{*}(first), et al., “Investigating Deep Learning Approaches for Human-Robot Proxemics”, IEEE International Symposium on Robot and Human Interactive Communication, 2018. (RO-MAN 2018)
- [51] **Yuan Gao**^{*}(first), et al., “When Robot Personalisation Does Not Help: Insights from a Robot-Supported Learning Study”, IEEE International Symposium on Robot and Human Interactive Communication, 2018. (RO-MAN 2018)
- [52] **Yuan Gao**^{*}(first), et al., “Human-Robot Proxemics Using Recurrent Neural Networks”, IEEE International Symposium on Robot and Human Interactive Communication, 2018. (RO-MAN 2018)
- [53] Peng Zhang, **Yuan Gao**, et al., “Less is More: Learning More with Concurrent Transmissions for Energy-Efficient Flooding”, Proceedings of the EAI International Conference on Mobile and Ubiquitous Systems, 2017. (MobiQuitous 2017)
- [54] Mohammad Obaid, **Yuan Gao**, et al., “Exploring Users’ Reactions Towards Tangible Implicit Probes for Measuring Human-Robot Engagement”, International Conference on Social Robotics, 2017, 402-412. (ICSR 2017)
- [55] **Yuan Gao**^{*}(first), et al., “Personalised Human-Robot Co-Adaptation in Instructional Settings Using Reinforcement Learning”, IVA Workshop on Persuasive Embodied Agents for Behavior Change, 2017. (PEACH 2017)
- [56] **Yuan Gao**^{*}(first), Dorota Glowacka, “Deep Gate Recurrent Neural Network”, Asian Conference on Machine Learning, 2016, 350-365. (ACML 2016)
- [57] **Yuan Gao**^{*}(first), et al., “OfficeHours: A System for Student Supervisor Matching Through Reinforcement Learning”, Companion Proceedings of the International Conference on Intelligent User Interfaces, 2015. (IUI Companion 2015)
- [58] Chuang Wang, Zelin Wang, Yingzhe Li, Chao Yu, Jianjun Luo, Zhanxia Zhu, Tin Lun Lam, **Yuan Gao**[†], “O-CAT: Three-Dimensional Orbital Capture with a Tethered Net via Multi-Agent Deep Reinforcement Learning”, IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2026. (submitted, T-SMCS)
- [59] Wei Yuan, Fengwen Liu, Ruize Wei, Zongwei Wang, **Yuan Gao**, Hu Huang, Binxing Fang, Qing Liao, “Safety of Vision-Language-Action Models: A Survey from Lifecycle Perspectives”, Authorea Preprints, 2026. (submitted, TPAMI)
- [60] Daicheng Song, Peng Ju, Kaiwen Yao, **Yuan Gao**[†], “E2I-Mem: Role-Specific Experience-to-Insight Memory for Heterogeneous Multi-Robot Urban Inspection”, IEEE International Conference on Robotics and Automation, 2027. (submitted, ICRA 2027)
- [61] Wenqiang Lai, **Yuan Gao**[†], et al., “Efficient Online Task-Oriented Visual Self-Modeling for 3D Motion Planning”, IEEE International Conference on Robotics and Automation, 2027. (submitted, ICRA 2027)